

Research Paper

A numerical study on the performance of chemisorption heat pump according to various design conditions

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ABSTRACT

To achieve net zero, research on waste heat recovery systems is actively underway, and interest in chemisorption heat pumps is also increasing. In a chemisorption heat pump, the performance increases as the temperature of the heat source increases. However, in order to recover a larger amount of waste heat, it is necessary to develop a system that can utilize waste heat at a lower temperature. Most existing studies on chemisorption heat pumps have been conducted on systems operated using heat sources above 65 °C, and most studies have presented system performance results at specific design points. In this study, changes in system performance according to system design were quantitatively analyzed for a system driven by a heat source at 40 °C, which is lower than the existing literature studies. A 1-D transient analysis model was developed and an analytical study was conducted. Performance analysis was conducted according to pressure drop requirement, number of tubes inside the heat exchangers and reactors, and number of fins inside the reactor, and the optimal design point for each condition was suggested. The cooling capacity, coefficient of performance, specific cooling power at the optimal design point are 1100.6 W, 0.3, 45.7 W/kg respectively.

1. Introduction

Net zero is a hot topic worldwide, and various technologies are being developed to achieve it. Globally, more than 40 countries including South Korea [1] have made pledges to go net zero. As of 2020, 33.9 Gt of CO₂ is emitted globally, of which 2.9 Gt comes from buildings. About half of the energy consumed in a building is used for heating, cooling, and hot water [2]. Therefore, developing technologies that can reduce the carbon emissions from heating and cooling can contribute to carbon neutrality. Using waste heat to operate a thermal-driven heat pump system can reduce carbon emissions from heating and cooling.

Absorption and adsorption heat pumps are representative thermal-driven heat pump systems. For most studies on these two systems, the temperature of the thermal energy to drive the system is around 70 °C or higher. In a chemisorption heat pump, the highest temperature heat source is supplied to the desorption reactor. In the desorption reaction, the higher the temperature, the higher the desorption rate and amount. Therefore, the performance of a chemisorption heat pump increases as the temperature of the heat source supplied to the desorption reactor increases when the temperatures of the secondary fluids supplied to the other components are the same. Therefore, if a system driven by a heat

source with a lower temperature compared to existing literature is developed, it can be applied to heat source conditions with a higher temperature than this heat source. Therefore, it is important to develop a system that can be driven by a lower temperature heat source, and in most existing studies, the temperature of the heat source supplied to desorption reactor was above 65 °C. Therefore, this study was conducted on systems that can be driven by lower temperature heat sources. Through this, there will be a new opportunity to utilize the low-temperature waste heat that would otherwise be wasted. Z. Yang et al. [3] reviewed ammonia-based chemisorption heat pumps and summarized chemisorption reactions between ammonia and metal halide salts. Z. Yang et al. [3] showed that adsorbents can be operated on a 40 °C heat source. This result shows that low-temperature heat sources can operate chemisorption heat pumps, depending on the choice of working fluid and adsorbent.

Many studies have been conducted on chemisorption heat pumps from various perspectives. Z. Q. Jin et al. [4] measured the thermal conductivity and permeability of adsorbents for the sorption process. In their study, the thermal conductivity and permeability varied with the ratio of expanded graphite and the density of the adsorbent, and the optimal conditions should be derived accordingly. Jong Hun Han et al. [5] reported the effective thermal conductivity and permeability of the

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Nomenclature		z	Length of axial direction [m]
A	Cross section area [m ²]	<i>Greek letters</i>	
Ar	Pre-exponential factor [-]	ρ	Density [kg/m ³]
C	Specific heat [J/kg.K]	θ	rotation angle [rad]
COP	Coefficient of performance [-]	φ	Heat transfer rate per area [W/m ²]
Cond	Condenser	<i>Subscripts</i>	
dP	Pressure drop [Pa]	2nd	Secondary
Ea	Activation energy [-]	ads	Adsorbent
Eva	Evaporator	ADS	Adsorption
HX	Heat exchanger	cond	Condenser
h	Enthalpy [J/kg]	conv	Convective
k	Thermal conductivity [W/m.K]	des	Desorption reaction
m	Mass [kg]	DES	Desorption
O.D.	Outer diameter [m]	eq	Equilibrium
P	Pressure [Pa]	EG	Expanded graphite
q	Heat transfer rate per unit volume [W/m ³]	eva	Evaporator
$\dot{q}$	Heat transfer rate per unit length [W/m]	fric	Friction
Q	Heat transfer rate [W]	i	Inner
r	Radius [m]	ins	Insulation
R	Universal gas constant [J/K.mol]	m	metal
SCP	Specific cooling power [-]	nom	Nominal
T	Temperature [K]	o	Outer
t	Time [second]	p	Pressure
u	Flow velocity [m/s]	r	Reaction
w	Mass flow rate [kg/s]		
x	Conversion ratio [-]		

graphite-MnCl₂ complex and conducted numerical simulations based on the measured properties. Asif Sha A. et al. [6] developed a new type of adsorbent based on activated carbon, graphite power, and metal organic framework. Adsorption/desorption performance tests and thermal conductivity were measured for the newly developed adsorbent. Z. Yang et al. [3] and L. W. Wang et al. [7] reviewed and summarized adsorption working pairs for ammonia. Both studies showed the reaction enthalpy and reaction entropy of each chemisorption reaction. Josée Trudel et al. [8] experimented to measure the equilibrium data of an ammonia-CoCl₂ system under pressure greater than 1 atm. Z. S. Zhou et al. [9] mentioned that actual refrigeration processes are non-equilibrium processes and that analyzing system performance based on equilibrium conditions can cause errors. Then, Z. S. Zhou et al. [9] measured and suggested a non-equilibrium curve of MnCl₂-ammonia and CaCl₂-ammonia working pairs. These studies are related to the fundamental experiments needed to design and analyze chemisorption heat pump systems.

Other researchers have used this information to research cycle design and analyze the performance of chemisorption systems. Huashan Bao et al. [10,11] conducted cycle analysis and proposed an optimized chemisorption cycle for power generation [10] and integrated chemisorption cycles for heat recovery and energy storage. Zhiyao Yang et al. [12] compared different working pairs and cycle configurations of ammonia chemisorption heat pumps for the heating mode. T. X. Li et al. [13] reviewed and summarized a solid-gas sorption refrigeration cycle. They showed different forms of thermodynamic cycles. Hua-Jiang Huang et al. [14] developed a gas-solid chemisorption heat pump model and compared the model results with that of experiments. M. V. Oro et al. [15] performed modeling of a chemisorption heat pump system using ammonia-NaBr to recover the waste heat of a fuel cell. Zhiyao Yang et al. [16] conducted dynamic modeling and performance analysis of a chemisorption heat pump for cold climates. They analyzed the performance based on daily operations and presented the heating COP. A. R. Pachó [17] developed a carbon-ammonia adsorption heat pump model and compared the analytical and experimental results. Huashan Bao et al. [18] developed a model for a chemisorption refrigeration and

power cogeneration system. They considered a system using ammonia-MnCl₂. They compared the results of the component-level analysis with experimental results. V. Baiju et al. [19] presented transient model of adsorption cooling system. They considered the system with activated carbon – ethanol. They suggested changes in system performance depending on operating conditions under specific design conditions. They conducted research on systems operated by heat sources ranging from 65 to 100 °C. S. Himmers et al. [20] conducted experiment and suggested non-equilibrium line. Based on the experimental results, they presented modeling of ammonia sorption reactor and compared simulation results with experimental results. They focused on the single reactor. G. H. Atkinson et al. [21] conducted experimental and numerical study on the resorption heat pump system. They manufactured a reactor by fixing an adsorbent in a hexagonal shape on the outer wall of the tube and performed an analysis on this shape. H. Hu et al. [22] developed simulation model of chemisorption heat pump with ammonia and SrCl₂. They suggested control method of the system. X. Xie et al. [23] suggested model for hybrid sorption-compression heat pump system. A system that produces hot water around 100 °C using waste heat of 50 to 70 °C was presented. R. G. Oliveira and D. J. Generoso [24] conducted experiments on the performance of a chemisorption chiller driven by hot water between 65 °C and 80 °C under various operation conditions. In their study, the COP (Coefficient of performance) and SCP (Specific cooling power) were about 0.32–0.36 and 46.6–64.1 W/kg, respectively. J. Xu et al. [25] studied a resorption system, using ammonia as a working fluid. NH₄Cl and MnCl₂ were utilized as a LTS (Low temperature salt) and HTS (High temperature salt), respectively. R. Wang et al. [26] and L. Wang et al. [27] wrote textbooks about adsorption technology. These textbooks contain detailed descriptions and specific examples of adsorption systems. S. Himmers and R. E. Critoph [28] modeled chemical heat transformations. G. H. Atkinson et al. [29] suggested the sorption characteristics of NH₄Cl and ammonia for a heat pump. G. H. Atkinson et al. [30] developed a resorption heat pump and conducted experiments. NaBr and MnCl₂ were used as low and high temperature salts, respectively. H. S. Kim et al. [31] designed a 100 W

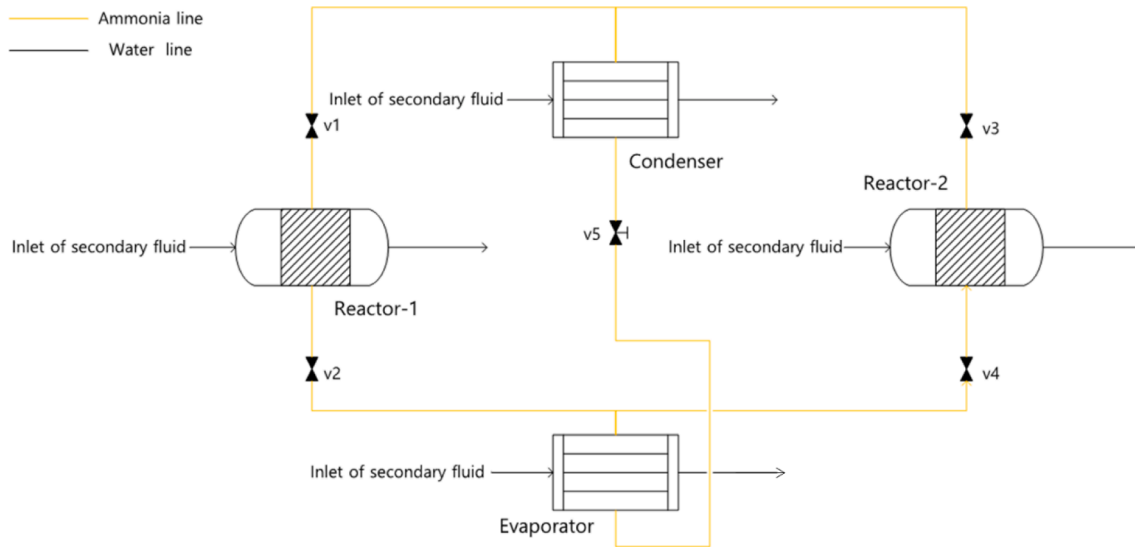


Fig. 1. Schematic diagram of a chemisorption heat pump system.

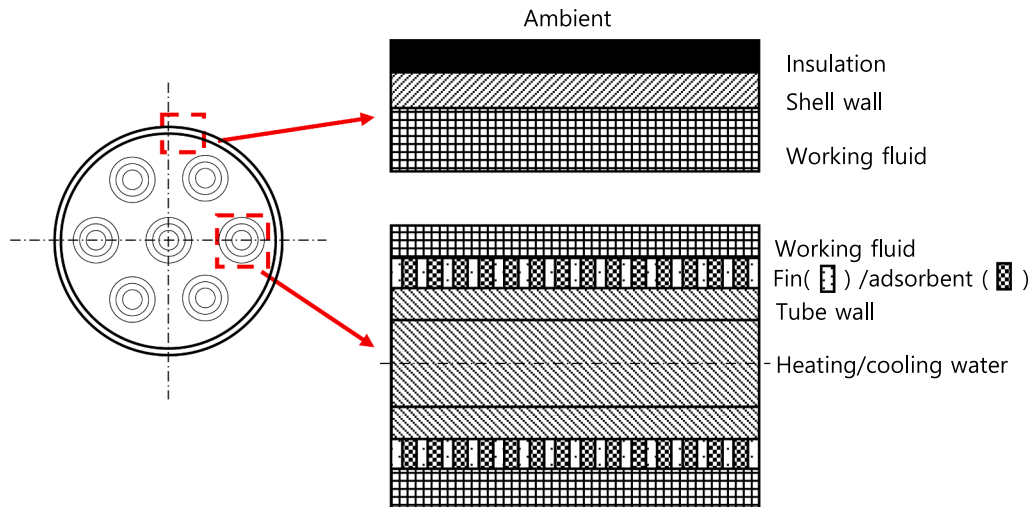


Fig. 2. Drawing of the reactor and schematic of the reactor for modeling [26].

class chemisorption heat pump and conducted experiments. They used a heat source of 40 °C to operate the chemisorption heat pump system. The COP and SCP were about 0.2 and 87 W/kg, respectively. H. S. Kim et al. [32] modeled a chemisorption heat pump system that used ammonia and NaBr as the working pair. They suggested some kinetic coefficients for the reaction and compared the simulation and experimental results.

As summarized above, studies have been conducted on working pair, cycle design, system modeling, and experiments have been done on the system performance of chemisorption heat pumps. However, the optimal design of chemisorption heat pump systems has not been investigated. This work conducted a numerical study on the optimal

design of a chemisorption heat pump system driven by a heat source of 40 °C. For a chemisorption heat pump system consisting of a finned-tube reactor, a flooded evaporator, and a surface condenser, this study analyzed the system performance according to the design parameters of each component. It proposed an optimal design based on the results.

2. System description

In this paper, a 2-bed system was considered. Fig. 1 shows a schematic diagram of the system. As described in Fig. 1, the system consists of two reactors, an evaporator and a condenser. Previous research has suggested many different types of reactors [14,18,20,21,24]. The most widespread of these reactor forms is the finned-tube reactor. This type of reactor is achieved by packing the adsorbent between the fins of a finned tube shown in Fig. 2. Expanded graphite impregnated with NaBr is considered as adsorbent which is the same adsorbent with previous study [32]. The mass ratio of expanded graphite and NaBr is 1:2 and the average density of adsorbent is 244 kg/m³.

Each component should have a secondary heat source to operate the system. In this study, the inlet side conditions of each component secondary fluid were set as the boundary condition. The desorption reactor should have a high temperature secondary fluid, the adsorption reactor

Table 1
Operation modes and valve positions for each mode.

Mode	Valve No.				
	1	2	3	4	5
(a) Adsorption(Rx1)/Desorption(Rx2)	X	O	O	X	X
(b) Preparation	X	X	X	X	O
(c) Adsorption(Rx2)/Desorption(Rx1)	O	X	X	O	X
(d) Preparation	X	X	X	X	O

and condenser with a medium temperature, and the evaporator with a low temperature.

To operate a chemisorption heat pump continuously, alternating operations should be performed. To perform alternating operations, four operating modes were defined, and in each operating mode, valves 1–5 were opened and closed shown in Table 1. Modes (a) and (c) are adsorption and desorption operations, and modes (b) and (d) are preparation operations for modes (a) and (c). In mode (a), Reactor 1 is connected to the evaporator, and Reactor 2 is connected to the condenser. The working fluid evaporated from the evaporator travels to Reactor 1, where the adsorbent inside Reactor 1 adsorbs it. On the adsorbent in reactor 2, the working fluid desorbs and moves to the condenser, where it condenses. Therefore, in mode (a), reactor 1 acts as the adsorption reactor and reactor 2 as the desorption reactor. Mode (c) is the reverse of mode (a), with the reactor working in reverse. In modes (a) and (c), the working fluid in the evaporator is supplied thermal energy from the secondary fluid, and the secondary fluid of the evaporator is cooled in this process. In modes (b) and (d), the inlet temperature of the secondary fluid fed to each reactor is changed instantaneously to change the role of the reactor. In addition, valve 5 is opened, and the working fluid in a liquid state, which was condensed in the condenser in the previous mode, moves to the evaporator. This process maintains the amount of refrigerant inside the evaporator and condenser. Modes (a) ~ (d) are repeated, and alternating operations are performed, which enables continuous operation of the chemisorption heat pump.

3. Modeling of a chemisorption heat pump system

Developing a transient model to simulate the operation mentioned above is necessary. To develop the model, it is necessary to model and integrate reactors, condensers, evaporators, and valves. The main components were modeled by referring to a study by H. S. Kim [32]. In this study, a transient model was developed and analyzed using DYMOLA 2022. It used Cvode, one of the solvers provided by Dymola, to perform the analysis, and the interval length and tolerance were set to 1 s and $1e^{-6}$, respectively.

As shown in Fig. 2, the secondary fluid flows inside the tube, and an adsorption (or desorption) reaction occurs on the adsorbent outside the tube. To simulate this, it is necessary to consider the heat transfer at the fluid inside the tube, the tube outer wall, the fins and adsorbent section, the fluid on the shell side, the shell outer wall, and the atmosphere. At the same time, it should reflect the working fluid reaction on the adsorbent. The actual structure of the reactor is 3-D, but in system analysis, many previous studies [15,17,18,32] performed a 1-D analysis based on some assumptions. In this study, 1-D modeling was conducted, and the following assumptions were applied:

- A fully developed and axisymmetric flow is inside the tube.
- The pressure and enthalpy of the working fluid of the shell side are uniform.
- The pressure and enthalpy of the secondary fluid of the tube side are uniform along the radial direction.
- Heat transfer in the axial direction is not considered.
- Heat and mass transfers in the radial direction are constant with the angle.

Based on these assumptions, the following mass, momentum and energy equations for the secondary fluid are used.

Mass equation:

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial z} = 0 \quad (1)$$

Momentum equation:

$$\frac{\partial(\rho u^2)}{\partial z} = -\frac{\partial P}{\partial z} - \frac{\partial P}{\partial z} \bigg|_{\text{fric}} \quad (2)$$

Table 2

Reaction coefficients and thermal properties of the tube, fin, adsorbent and insulation.

Parameter	Value	Parameter	Value
Thermal conductivity of tube (k_m) [33]	14.9 [W/(m•K)]	Ar (Adsorption) [15]	17.1 [1/s]
Density of tube (ρ_m) [33]	7900 [kg/m ³]	Ea (Adsorption) [15]	4200 [J/mol]
Specific heat of tube (c_m) [33]	477 [J/(kg•K)]	Ar (Desorption) [15]	2e5 [1/s]
Thermal conductivity of fin (k_m) [33]	14.9 [W/(m•K)]	Ea (Desorption) [15]	42281 [J/mol]
Thermal conductivity of adsorbent (k_{ads}) [15]	0.43 [W/(m•K)]	M (Adsorption) [32]	4
Thermal conductivity of insulation (k_{ins}) [33]	0.038 [W/(m•K)]	N (Adsorption) [32]	2
Density of insulation (ρ_{ins}) [33]	32 [kg/m ³]	M (Desorption) [32]	1
Specific heat of insulation (c_{ins}) [33]	835 [J/(kg•K)]	N (Desorption) [32]	1

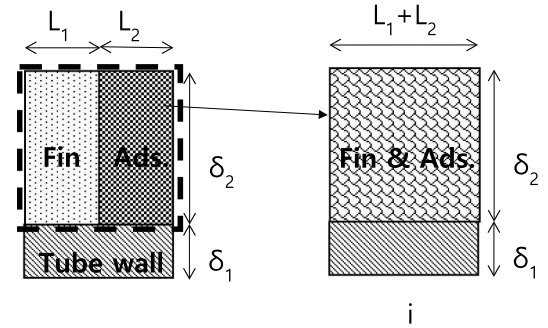


Fig. 3. Simplified schematic of the fin and adsorbent for 1-D modeling [32].

Energy equation:

$$\frac{\partial(\rho h)}{\partial t} + \frac{\partial(\rho u h)}{\partial z} = q_{conv} + \frac{\partial P}{\partial t} \quad (3)$$

In Eq. (3), to calculate the convective heat transfer (q_{conv}), a heat transfer coefficient is required, and in this study, the Gnielinski correlation [33] equation is applied for the turbulent conditions, and a constant Nu value of 4.36 is applied for laminar flow conditions. In Eq. (2), friction loss is simply treated as the following equation.

$$\frac{\partial P}{\partial z} \bigg|_{\text{fric}} = \frac{\frac{\partial P}{\partial z} \bigg|_{\text{fric, nom}}}{w_{nom}} \bullet w \quad (4)$$

The tube wall was modeled based on the following energy equations:

$$\rho c_m A_m \frac{dT}{dt} = 2\pi r \varphi_i + 2\pi r \varphi_o + \dot{q}_{reaction} \quad (5)$$

$$\varphi_i = \frac{k_m}{r_i \ln\left(\frac{r_i + r_o}{2r_i}\right)} (T_i - T_m) \quad (6)$$

$$\varphi_o = \frac{k_m}{r_o \ln\left(\frac{2r_o}{r_i + r_o}\right)} (T_o - T_m) \quad (7)$$

The thermal properties of the tube, fin and adsorbent are shown in Table 2. The same Eqs. (5) to (7) were applied to the fin/adsorbent section as to the tube. The thermal conductivity in this section was applied as the average thermal conductivity of the fin and adsorbent (Fig. 3), and the average thermal conductivity was calculated and applied in the same way as the method proposed by H. S. KIM [32]. The

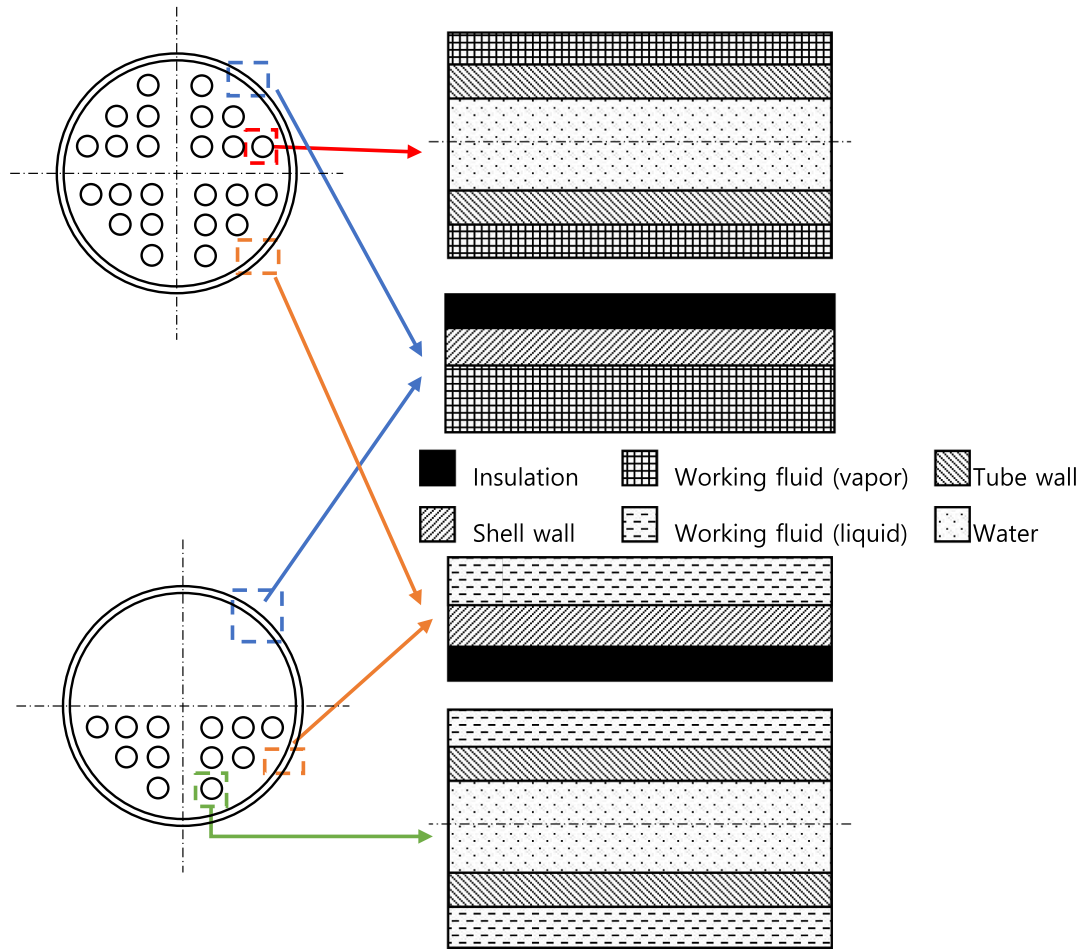


Fig. 4. Drawing of the heat exchangers and schematic of the heat exchangers for modeling [32].

heat capacity of this section is obtained from Eq. (8).

$$(mc_P)_{fin/adsorbent} = (mc_P)_{EG} + (mc_P)_{NH_3} + (mc_P)_{NaBr} + (mc_P)_{NaBr \cdot 5.25NH_3} \quad (8)$$

Adsorption and desorption rates were calculated based on the Eqs. (9) to (12).

$$\begin{aligned} \frac{dx}{dt} &= c^* A \exp\left(-\frac{E_a}{RT}\right) f(x)^M (\Delta P_{eq})^N \quad (c^* = 1 \text{ for adsorption, } c^* \\ &= -1 \text{ for desorption}) \end{aligned} \quad (9)$$

$$f_{ADS}(x) = 1 - x \text{ or } f_{DES}(x) = x \quad (10)$$

$$\Delta P_{eq,ADS} = \frac{P_{eva} - P_{eq}}{P_{eq}} \text{ or } \Delta P_{eq,DES} = \frac{P_{eq} - P_{cond}}{P_{eq}} \quad (11)$$

$$\frac{P_{eq}}{P_0} = \exp\left(\frac{-\Delta H_r}{RT} + \frac{\Delta S_r}{R}\right) \quad (12)$$

The constants required in Eq. (9) were based on research results from existing literatures, which are described in Table 2. The ΔH_r (21030.86 J/mol) was obtained from previous study [32]. The ΔS_r was calculated by eq. (13) which is also suggested by H. S. Kim et al. [32].

$$\Delta S_r = 89.21 + 0.3255T_1 - 0.3012T_2 - 8.333e^{-4}T_1^2 - 8.554e^{-4}T_1T_2 + 1.511e^{-3}T_2^2 \quad (13)$$

In the case of the working fluid in the shell side, Eqs. (1) and (3) were applied. However, when applying Eq. (3), q_{conv} considers the heat transfer from the fin/adsorbent section and the shell wall. The same Eqs. (5) to (7) were applied to the shell wall and insulation layer as to the

tube.

Z. Yang et al. [3] summarized the equilibrium lines of the chemisorption reaction between ammonia and metal halide salts. They showed that among various metal halide salts, NH_4Cl , $PbCl_2$ and $NaBr$ are suitable for a reaction under a low-temperature condition. R. G. Oliveira and D. J. Generoso [24] experimented using ammonia and $NaBr$, where the temperature of a high-temperature heat source was 70 °C. H. S. Kim et al. [31,32] also experimented using ammonia and $NaBr$ when the temperature of the heat source was 40 °C. Because previous studies have shown that the ammonia- $NaBr$ combination can react under low temperature conditions, the ammonia- $NaBr$ reaction was modeled in this study. When a chemical reaction occurs, the rate of the chemical reaction can be calculated based on the equations and reaction parameters presented by H. S. Kim et al. [32]. In this study, chemical reaction modeling was conducted as proposed by H. S. Kim et al. [32].

Fig. 4 shows the design of the evaporator and condenser. The evaporator has tubes immersed in a liquid refrigerant, and the condenser has tubes between a gaseous refrigerant. Modeling of the evaporator and condenser is the same as the modeling of the reactor, except that there is no fin/adsorbent section. When applying the shell-side energy equation for the heat transfer coefficient at the tube wall, it used the film condensation correlation [34] for the condensation process, the Cooper correlation [35] for the evaporation process, and the Churchill and Chu correlation [33] for the natural convection process.

When modeling was performed in the above manner, the comparison between the simulation and the experimental results was conducted by H. S. Kim et al. [32]. The results are shown in Fig. 5 [32]. As can be seen in Fig. 5, it can be seen that similar results to the experiment can be

derived through the simulation model.

4. Results and discussion

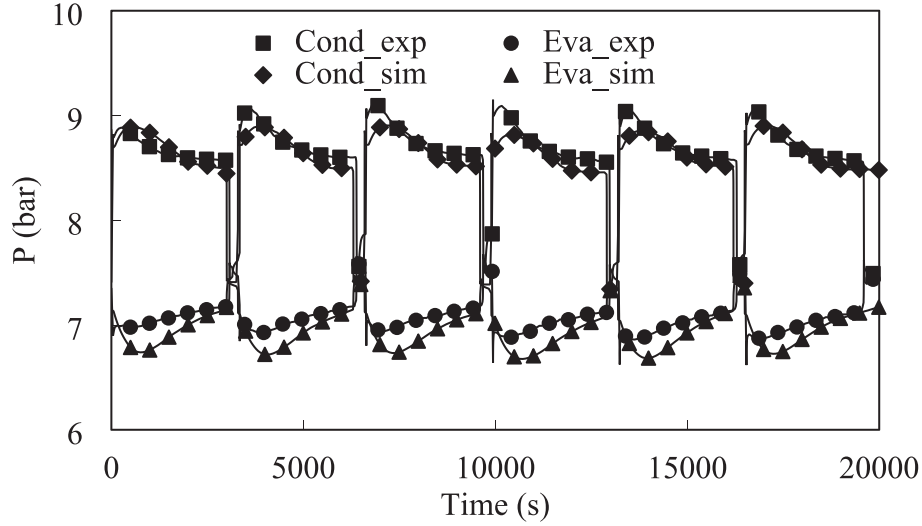
4.1. Data reduction

The performance of the chemisorption heat pump is affected by the operating time for each of the operating modes shown in Table 1. In this study, the operation time (t_1) for modes (a) and (c) was set to 50 min, and the operation time (t_2) for modes (b) and (d) was set to 5 min, and then, the performance analysis was performed. Fig. 6 shows the analysis results under specific design conditions, and it can be seen that the outlet temperatures of the evaporator and condenser secondary fluids repeat in

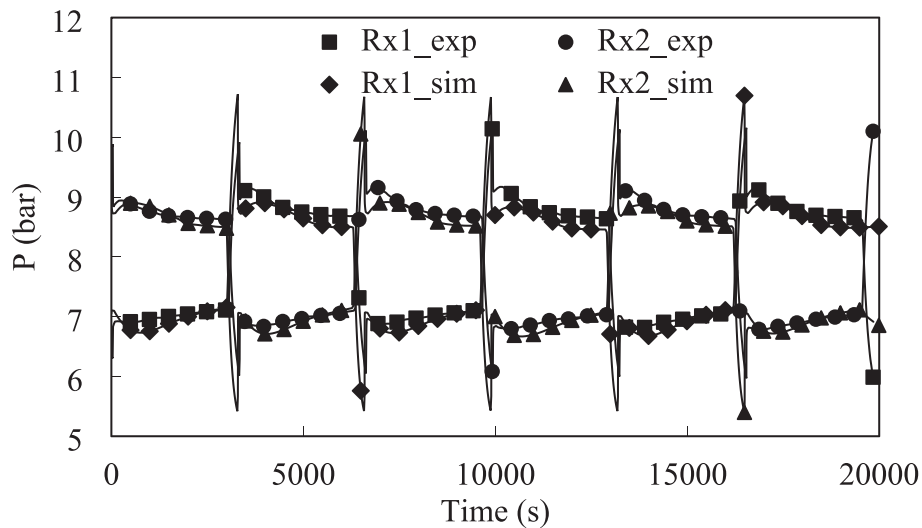
a specific pattern during modes (a) through (d). The cooling capacity of the system is defined as the time-averaged heat input to the evaporator during modes (a) through (d), and the system COP is defined as the cooling capacity divided by the time-averaged heat input to the desorption reactor. The SCP is defined as the cooling capacity divided by the mass of the adsorbent charged in the whole system. The cooling capacity, COP and SCP can be calculated by the following equations.

$$Q_{eva} = \frac{\sum_{t=0}^{t=2(t_1+t_2)} [w_{eva,2nd} C_p (T_{eva,2nd,inlet} - T_{eva,2nd,outlet})]}{2(t_1 + t_2)} \quad (14)$$

$$Q_{des} = \frac{\sum_{t=0}^{t=2(t_1+t_2)} [w_{des,2nd} C_p (T_{des,2nd,inlet} - T_{des,2nd,outlet})]}{2(t_1 + t_2)} \quad (15)$$

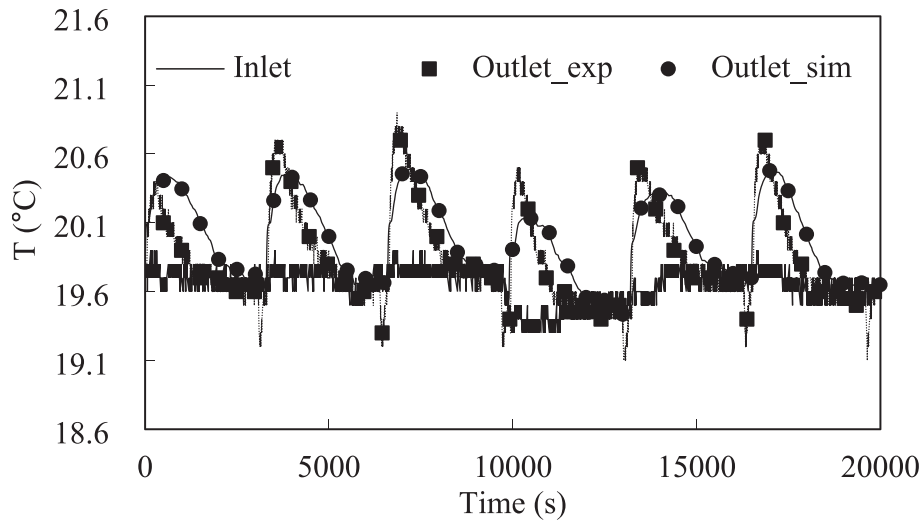


(a) Pressure in heat exchangers

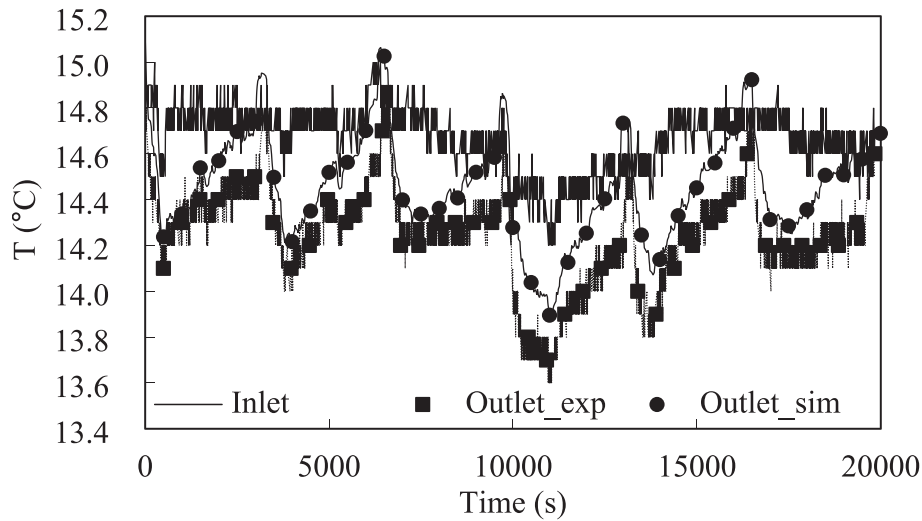


(b) Pressure in reactors

Fig. 5. Comparison between simulation and experimental results of H. S. Kim et al. [32].



(c) Inlet and outlet temperature of secondary fluid in condenser



(d) Inlet and outlet temperature of secondary fluid in evaporator

Fig. 5. (continued).

$$COP = \frac{Q_{eva}}{Q_{des}} \quad (16)$$

$$SCP = \frac{Q_{eva}}{m_{adsorbent}} \quad (17)$$

These are equivalent to the cooling capacity, COP and SCP defined in a study by R. G. Oliveira and D. J. Generoso [24].

H. S. Kim et al. [32] reported on the characteristics and system performance of a fin-tube inside a reactor using the tube diameter and fin height. They presented the cooling capacity, adsorbent loading in the system, and COP according to the fin-tube diameter and fin height. The COP tended to increase with smaller tube diameters and more extended fin heights. The cooling capacity increases with smaller tube diameters,

but there is a fin height at which the cooling capacity is maximized for each tube diameter.

4.2. Simulation condition

In this study, the system performance was analyzed according to the number of fin-tubes in the reactor, the number of tubes in the heat exchanger, the diameter of the heat exchanger tubes, the number of tube-passes in the reactor and the heat exchangers, and number of fins in the reactor. Table 3 shows the reactor and heat exchanger design parameters used in the analysis. Both reactors and heat exchangers are the shell-and-tube type, and the velocity of the secondary fluid supplied to the tube side affects the system performance. The velocity of the secondary fluid in the pipe is determined by the header design, which

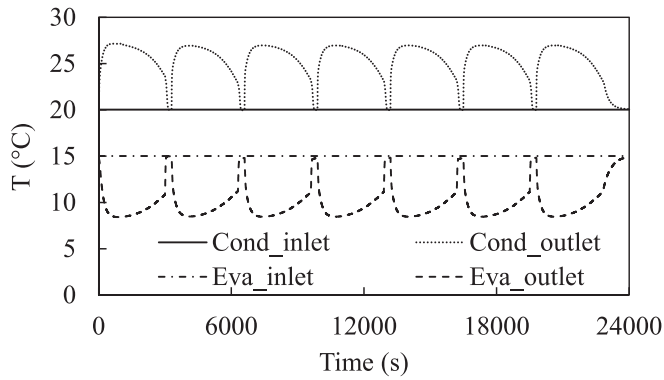


Fig. 6. Simulation results for specific design conditions

Table 3

The design parameters of the reactor and heat exchanger.

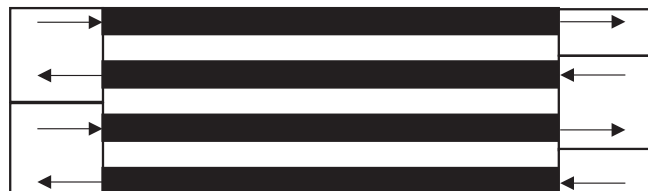
Parameter	Value
Shell outer diameter of reactor (mm)	380
Shell inner diameter of reactor (mm)	350
Tube outer diameter of reactor (mm)	9.53
Tube inner diameter of reactor (mm)	7.03
Fin height of reactor (mm)	6.0
No. of fins per meter (-)	276
No. of fin-tubes in reactor (-)	100, 120, 140, 160, 180, 200
Length of fin-tubes in reactor (mm)	1000
Shell outer diameter of heat exchanger (mm)	216.3
Shell inner diameter of heat exchanger (mm)	199.9
Tube outer/inner diameter of heat exchanger (mm)	9.53/7.03, 12.7/9.39, 15.88/12.57



1 Pass



2 Pass



4 Pass

Fig. 7. Flow direction according to the number of passes.

Table 4

Number of passes of tubes in the reactors and heat exchangers for each tube-side pressure drop condition.

Reactor				
No. of tubes			No. of Passes (0.1 bar/1 bar)	
100			5/20	
120			10/24	
140			10/20	
160			10/20	
180			12/30	
200			10/25	

O. D. (mm)	Evaporator		Condenser	
	No. of tubes	No. of Passes (0.1 bar/1 bar)	No. of tubes	No. of Passes (0.1 bar/1 bar)
9.53	20	5/20	40	10/20
	40	10/20	80	20/40
	60	15/30	120	30/60
	90	18/45	180	36/60
	120	30/60	240	48/80
12.7	10	5/10	20	10/20
	20	10/20	40	20/40
	30	15/30	60	30/60
	60	30/60	120	40/60
	90	30/90	180	60/90
15.88	10	10/10	20	20/20
	20	20/20	40	20/40
	30	30/30	60	30/60
	40	20/40	80	40/80
	60	30/60	120	60/120

typically takes into account the pressure drop to determine the number of passes on the tube side. Here, the number of passes is defined as shown in Fig. 7. For the same mass flow conditions, a higher number of passes increases the pressure drop and heat transfer coefficient because the velocity of secondary fluid in the pipe increases. If the pressure drop is not considered, simply increasing the number of passes is beneficial to the performance of the system because it increases the heat transfer coefficient. However, the pressure drop should be considered when designing the actual system. The number of passes to ensure that the tube-side pressure drop in the evaporator, condenser, and reactor is less than or equal to 0.1 bar and 1 bar, respectively, was calculated. The flow rates of the secondary fluid supplied to the reactor and heat exchanger were set to 15 and 3 LPM, respectively. In the evaporator, condenser, and reactor, the number of passes to ensure that the tube-side pressure drop is less than or equal to 0.1 bar and 1 bar, respectively, was calculated and is shown in Table 4. In most cases, the number of passes for pressures below 0.1 bar is lower than that for pressures below 1 bar, except for the conditions where the heat exchanger pipe diameter is 15.88 mm.

4.3. Simulation results and discussion

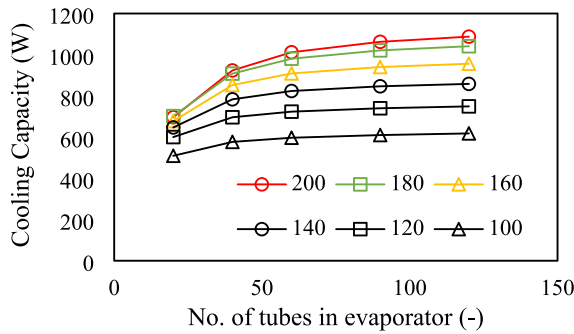
Fig. 8 shows the performance of the chemisorption heat pump system at conditions with a heat exchanger and reactor tube side pressure drop of 0.1 bar or less. Fig. 8(a – c), (d – f) and (g – i) show the analysis results for conditions with heat exchanger tube outer diameters of 9.53, 12.7, and 15.88 mm, respectively. In Fig. 8, each legend refers to the number of fin-tubes inside the reactor. The x-axis in Fig. 8 is the number of evaporator tubes, and the number of condenser tubes is twice as many as the number of evaporator tubes as shown in Fig. 4.

From Fig. 8(a), the average cooling capacity of the chemisorption heat pump system increases as the number of tubes in the heat exchangers increases. However, when the number of evaporator tubes is above 60, the performance becomes somewhat saturated, and the

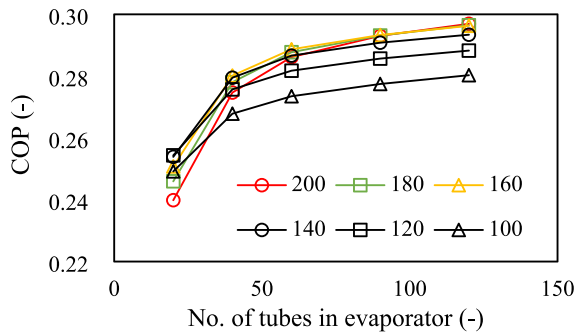
cooling capacity remains almost constant as the number of evaporator pipes increases. In the chemisorption heat pump system, COP is defined in Eq. (16), which is the cooling capacity relative to the sum of the heat input to raise the temperature of the desorption reactor structure at the start of the 'Preparation' mode and the heat input for the desorption reaction in the 'Adsorption/desorption' mode. Here, for the same reactor design conditions, the amount of heat supplied to the reactor structure at the start of the 'Preparation' mode is almost constant, while the amount of heat supplied to the desorption reaction in the 'Adsorption/desorption' mode increases with the increasing cooling capacity. In a chemisorption heat pump, the amount of heat input to raise the temperature of the desorption reactor's structure in the 'Preparation' mode is heat loss. However, when the cooling capacity increases, the amount of this heat loss remains almost constant, but the portion in Eq. (16) decreases, thereby increasing system COP. Therefore, the increase in the

numerator term of Eq. (16) is larger than that in the denominator term, the COP is increased shown in Fig. 8(b).

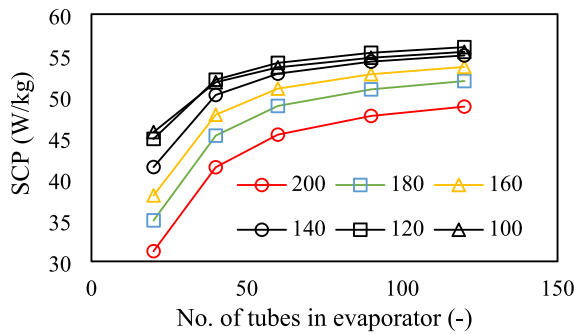
In the case of the reactor fin-tubes, the cooling capacity is almost saturated at around 180 fin-tubes; thus, increasing the number of fin-tubes here will not significantly increase the cooling capacity. For the same number of heat exchanger tubes, an increase in the number of reactor fin-tubes increases both the amount of heat supplied to the reactor structure at the start of the 'Preparation' mode and the amount of heat supplied for the desorption reaction in the 'Adsorption/desorption' mode. Therefore, the COP increases with an increase in the number of reactor fin-tubes only if the increase in the cooling capacity is greater than the increase in heat supplied to the reactor. In Fig. 8(b), the COP increases with the increase of the number of reactor fin-tubes in most cases for the condition of 100 to 160 reactor fin-tubes, which has a large increment in the cooling capacity with the increase of the number of



(a) Cooling capacity when the outer diameter of the tubes in the heat exchangers is 9.53 mm

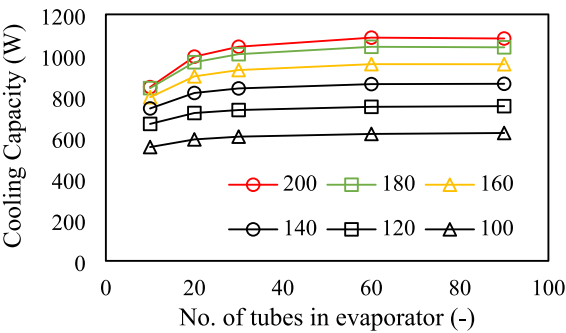


(b) COP when the outer diameter of the tubes in the heat exchangers is 9.53 mm

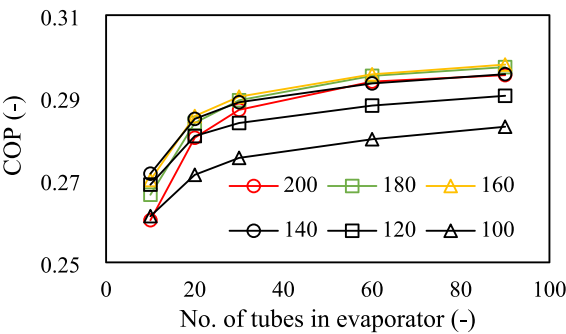


(c) SCP when the outer diameter of the tubes in the heat exchangers is 9.53 mm

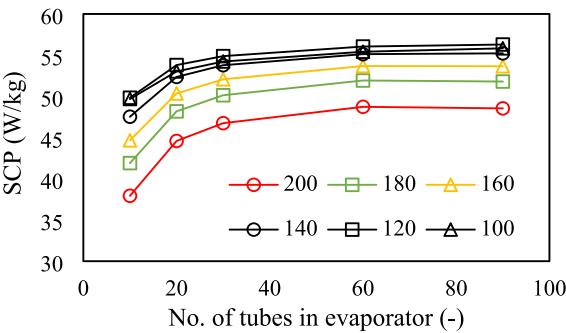
Fig. 8. Simulation results of the chemisorption heat pump when the pressure drops of the reactors and heat exchangers are 0.1 bar or less.



(d) Cooling capacity when outer diameter of tubes in heat exchangers is 12.7 mm

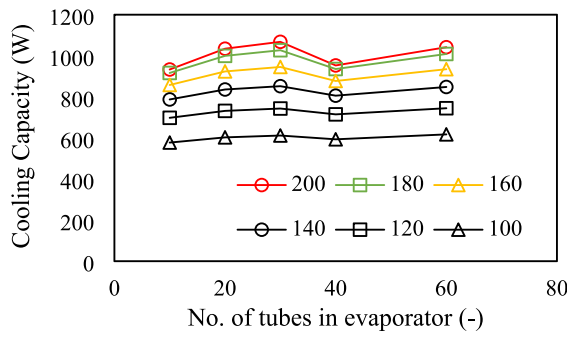


(e) COP when the outer diameter of the tubes in the heat exchangers is 12.7 mm

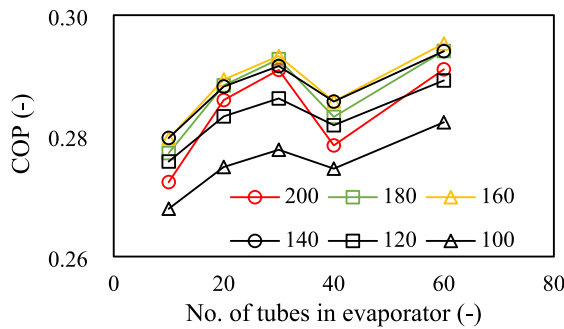


(f) SCP when the outer diameter of the tubes in the heat exchangers is 12.7 mm

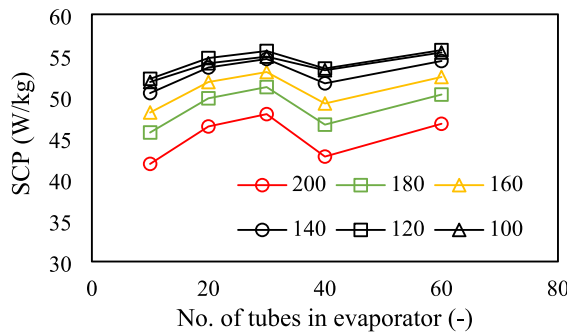
Fig. 8. (continued).



(g) Cooling capacity when the outer diameter of the tubes in the heat exchangers is 15.88 mm



(h) COP when the outer diameter of the tubes in the heat exchangers is 15.88 mm



(i) COP when the outer diameter of the tubes in the heat exchangers is 15.88 mm

Fig. 8. (continued).

reactor fin-tubes. In other areas, the COP decreases with the increase in reactor fin-tubes.

When the number of reactor fin-tubes is constant, the total adsorbent mass inside the reactor, which is the denominator of Eq. (17), is constant, and the SCP increases with the increase of the cooling capacity. Therefore, Fig. 8(c) shows that the SCP tends to increase with the increase of the number of heat exchanger tubes under the condition of a constant number of reactor fin-tubes, and similar to Fig. 8(a), the increase of the SCP starts to decrease when the number of evaporator tubes is 60. From Fig. 8(c), it can be seen that there is a number of reactor fin-

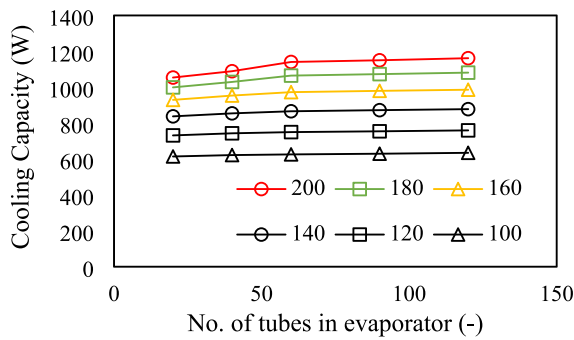
tubes that reach the maximum SCP when the number of the heat exchanger tubes is constant. Under the condition of the same number of heat exchanger tubes, the mass of the adsorbent in the system, which is the denominator in Eq. (17), changes in the same proportion as the number of reactor fin-tubes changes. As shown in Fig. 8(a), as the number of reactor tubes increases, cooling capacity increases but the amount of increase decreases. For example, when the number of evaporator tubes is 60 and the numbers of reactor tubes are 100, 120, and 140, the cooling capacity are 597.0, 723.7, and 823.9 W, respectively. That is, as the number of reactor tubes increases by 20 and 40 %, cooling

capacity increases by 21.2 and 38 %. As the number of reactor tubes increases, the mass of adsorbent charged into the system also increases in proportion. Therefore, comparing these three cases, the SCP is maximum when the number of reactor tubes is 120, and when the number of reactor tubes is increased or decreased from this value, the SCP decreases.

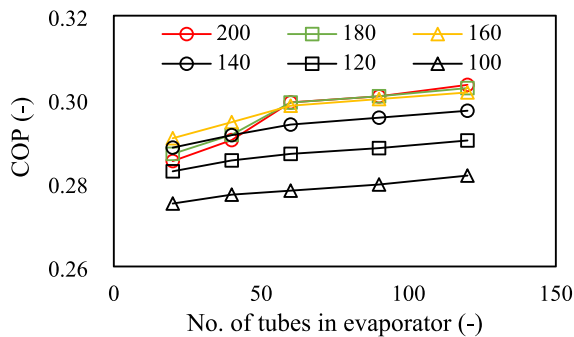
From Fig. 8(a – c), when the outer diameter of the heat exchanger is 9.53 mm, the system cooling capacity, COP, and SCP range from 509.4 to 1087.8 W, 0.24 to 0.296, and 31.2 to 55.4 W/kg, respectively, depending on the number of heat exchanger tubes and reactor fin-tubes. Looking at the cooling capacity changes with the increasing number of heat exchangers tubes and reactor fin-tubes, the optimal heat exchanger design point is when the numbers of evaporator and condenser tubes are 60 and 120, respectively, and the optimal number of reactor fin-tubes is

180. Under this condition, the cooling capacity, COP, and SCP are 980.7 W, 0.288, and 48.9 W/kg, respectively, which is not a significant performance difference from the maximum investment in the number of reactor fin-tubes and heat exchanger tubes.

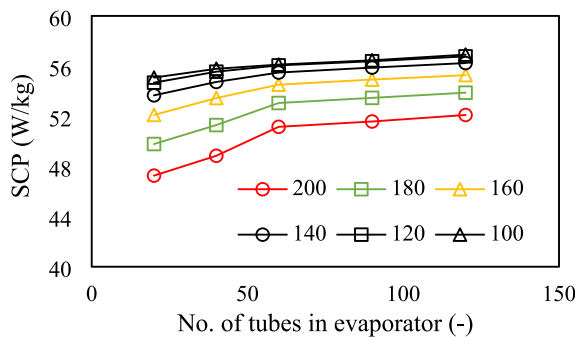
Fig. 8(d – f) show the analysis results for the case in which the heat exchanger tube diameter is 12.7 mm, showing a similar trend to the results in Fig. 8(a – c). In this condition, it can be seen that there is little improvement in system performance with more evaporator tubes when the number of evaporator tubes is above 30. This condition is the same as when the heat exchanger tube diameter is 9.53 mm, and it can be seen that there is little change in the performance with the increasing number of reactor fin-tubes when the number of reactor fin-tubes is above 180. As shown in Fig. 8(d – f), when the heat exchanger tube diameter is 12.7 mm, the cooling capacity, COP, and SCP can achieve performances



(a) Cooling capacity when the outer diameter of the tubes in the heat exchangers is 9.53 mm

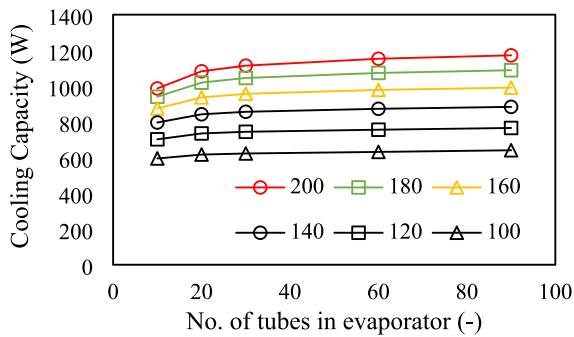


(b) COP when the outer diameter of the tubes in the heat exchangers is 9.53 mm

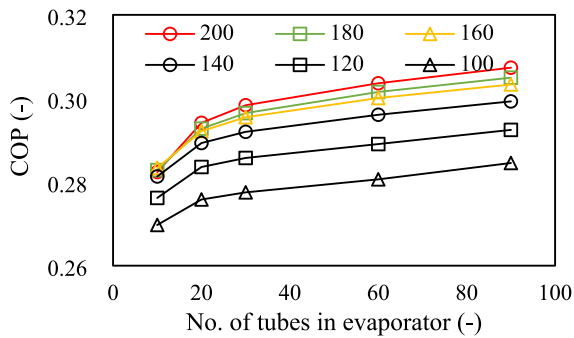


(c) SCP when the outer diameter of the tubes in the heat exchangers is 9.53 mm

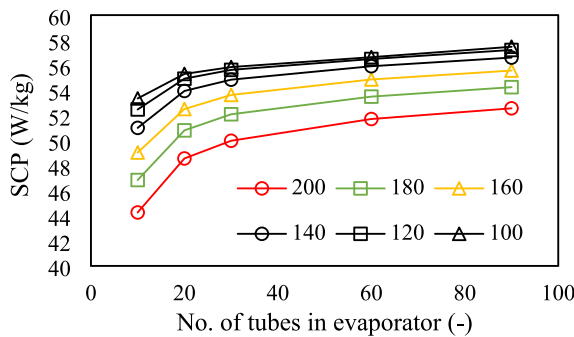
Fig. 9. Simulation results of the chemisorption heat pump when the pressure drops of the reactors and heat exchangers are 1 bar or less.



(d) Cooling capacity when the outer diameter of the tubes in the heat exchangers is 12.7 mm

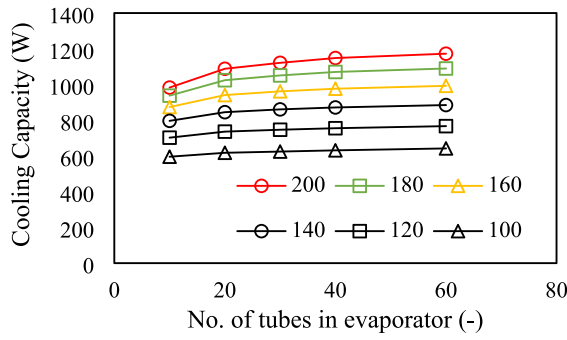


(e) COP when the outer diameter of the tubes in the heat exchangers is 12.7 mm

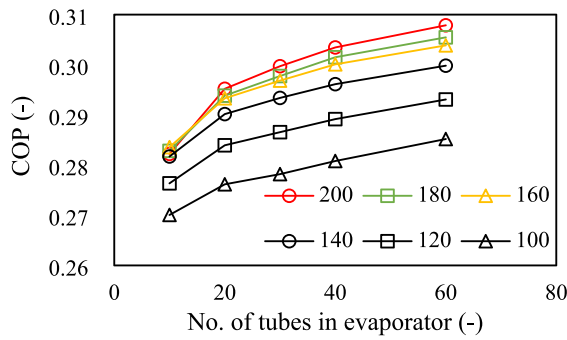


(f) SCP when the outer diameter of the tubes in the heat exchangers is 12.7 mm

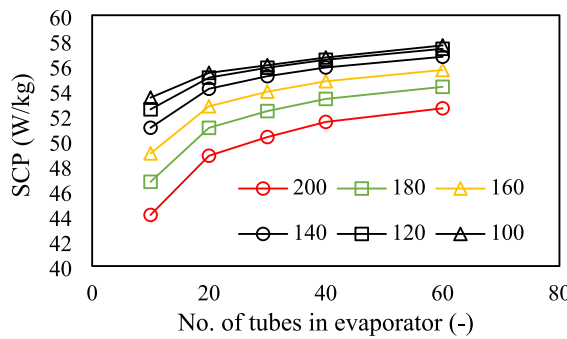
Fig. 9. (continued).



(g) Cooling capacity when the outer diameter of the tubes in the heat exchangers is 15.88 mm



(h) COP when the outer diameter of the tubes in the heat exchangers is 15.88 mm



(i) COP when the outer diameter of the tubes in the heat exchangers is 15.88 mm

Fig. 9. (continued).

ranging from 554.0 to 1086.8 W, 0.26 to 0.298, and 37.9 to 55.8 W/kg, respectively, depending on the heat exchanger and reactor design conditions. When the diameter of heat exchanger tube is 12.7 mm, the optimal design point is around 30, 60, and 180 for the number of inner tubes in the evaporator, condenser, and reactor, respectively. Under this condition, the cooling capacity, COP, and SCP are 1006.3 W, 0.29, and 50.1 W/kg, respectively.

Fig. 8(g – i) show the results for a heat exchanger tube with an outer diameter of 15.88 mm. For a heat exchanger tube diameter of 15.88 mm, the system performance decreases when the number of tubes in the

evaporator increases from 30 to 40. Table 4 shows that for an evaporator tube diameter of 15.88 mm and a pressure drop of 0.1 bar or less, the number of evaporator tube passes is 30 and 20 when the number of evaporator tubes is 30 and 40, respectively. This means that the number of passes decreases as the number of evaporator tubes increases to satisfy the heat exchanger tube side pressure drop condition, which means that the flow rate of secondary fluid in the evaporator tubes decreases as the number of evaporator tubes increases from 30 to 40. The Re numbers of the tube side in evaporator are 4455 and 2227 when the numbers of tubes are 30 and 40, respectively, indicating that the flow inside the

Table 5

Maximum and minimum cooling capacities, COPs, SCPs under each design condition in Figs. 8 and 9.

Design condition	Maximum/minimum cooling capacity (W)	Maximum/minimum COP (-)	Maximum/minimum SCP (W/kg)
dP <0.1 bar, O.D. of tube in HX = 9.53 mm	509.4–1087.8	0.24–0.296	31.2–55.4
dP <0.1 bar, O.D. of tube in HX = 12.7 mm	554–1086.8	0.26–0.298	37.9–55.8
dP <0.1 bar, O.D. of tube in HX = 15.88 mm	557.5–1067.9	0.268–0.295	41.9–55.7
dP <1 bar, O.D. of tube in HX = 9.53 mm	614.2–1161.8	0.275–0.302	47.3–56.9
dP <1 bar, O.D. of tube in HX = 12.7 mm	594.9–1172.2	0.270–0.307	44.3–57.5
dP <1 bar, O.D. of tube in HX = 15.88 mm	595.7–1172.5	0.270–0.308	44.1–57.6

Table 6

The design values and performance results at the optimal design points under each tube diameter condition.

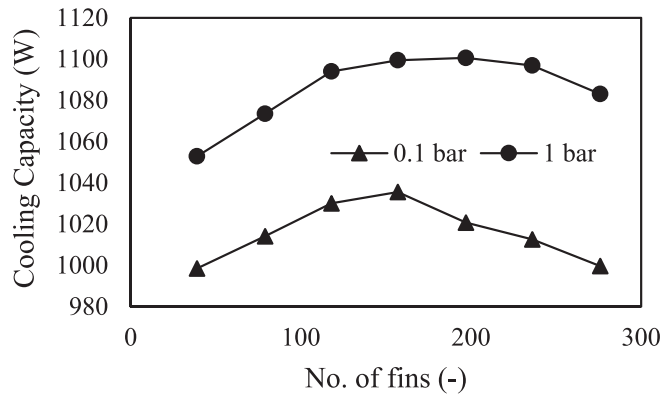
Case No.	Design condition	No. of tubes in evaporator/condenser/reactor (-)	Cooling capacity (W)	COP (-)	SCP (W/kg)	Mass of tubes in evaporator (kg)
1	dP <0.1 bar, O.D. of tube in HX = 9.53 mm	60/120/180	980.7	0.288	48.9	15.4
2	dP <0.1 bar, O.D. of tube in HX = 12.7 mm	30/60/180	1006.3	0.290	50.1	13.6
3	dP <0.1 bar, O.D. of tube in HX = 15.88 mm	20/40/180	999.6	0.288	49.8	11.7
4	dP <1 bar, O.D. of tube in HX = 9.53 mm	60/120/200	1140.8	0.300	51.2	15.4
5	dP <1 bar, O.D. of tube in HX = 12.7 mm	20/40/200	1083.1	0.291	48.6	9.1
6	dP <1 bar, O.D. of tube in HX = 15.88 mm	20/40/200	1088.1	0.295	48.8	11.7

evaporator tube changes from turbulent to laminar as the number of tubes increases from 30 to 40. Therefore, the heat transfer correlations that apply when the number of tubes is 30 and 40 are different. Thus, the difference in the heat transfer coefficients is calculated to decrease the overall UA, decreasing the system performance, even though the heat transfer area increases. From (g) to (i) in Fig. 8, it can be seen that the system performance does not increase significantly when the number of reactor tubes is above 180, which is similar to the other evaporator tube outer diameter conditions. For the evaporator and condenser inner tubes, installing more than 20 and 40 tubes, respectively, does not significantly improve performance. Under this condition, the cooling capacity, COP, and SCP are 999.6 W, 0.288, and 49.8 W/kg, respectively. When the heat exchanger tube outer diameter is 15.88 mm, the cooling capacity, COP, and SCP ranged from 577.5 to 1067.9 W, 0.268 to 0.295, and 41.9 to 55.7 W/kg, respectively, depending on the number of reactor and heat exchanger inner tubes. As shown in Fig. 8, in most cases, the system performance increases as the number of tubes in the heat exchanger increases. However, suppose the system is designed with the pressure drop condition of the secondary fluid already determined. In that case, it can be seen that in some cases, the system performance may decrease even if the number of tubes in the heat exchanger increases.

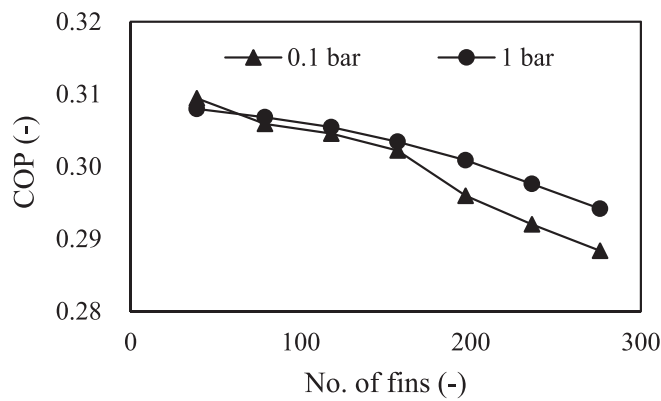
Fig. 9 shows the analysis results when the number of tube passes is set so that the pressure drops of the secondary fluid of the heat exchangers and reactor are 1 bar or less. Fig. 9(a–c), (d–f), (g–i) are the results when the tube outer diameters of the heat exchangers are 9.53, 12.7 and 15.88 mm, respectively. If the reactor has an inner diameter of 350 mm, the number of fin-tubes that can theoretically be installed inside the reactor is about 200. As shown in Fig. 9(a, d and g), if the heat exchanger is designed to have a pressure drop of 1 bar or less, the system performance increases almost directly with the number of fin-tubes inside the reactor, regardless of the inner tube diameter of the heat exchanger. The trend in Fig. 9 shows that the system performance improves as the number of heat exchanger tubes increases for all conditions. This is because the flow pattern at pressure drops below 1 bar is turbulent under all conditions.

Table 5 summarizes the values of Figs. 8 and 9. In case of

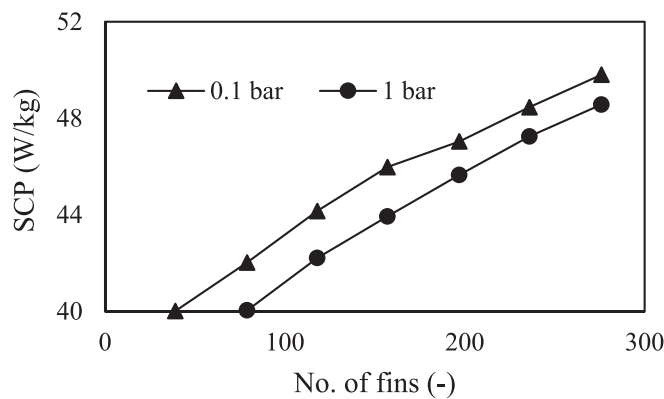
experimental results of R. G. Oliveira and D. J. Generoso [24], COP and SCP are 0.32–0.36 and 46.6–64.1 W/kg, respectively, under conditions where the temperature of the heat source is 65 to 80 °C. The analysis results of this study confirm that COP and SCP are 0.24–0.308 and 31.2–57.6 W/kg, respectively, depending on the design conditions. Although the heat source conditions for operating the system are different, it can be confirmed that the experimental results from existing literature and the analysis results of this study present a similar level of performance. An improvement in the performance can be seen when the pressure drop condition of the secondary fluid is 1 bar or less compared to 0.1 bar or less, when the tube diameter of the heat exchanger is 9.53 mm, the maximum values of the cooling capacity, COP, and SCP increase by 6.8, 2.0, and 2.7 %, respectively, as the pressure drop condition increases from 0.1 bar to 1 bar. When the heat exchanger has a tube diameter of 12.7 mm, the cooling capacity, COP, and SCP increase by 7.9, 3.0, and 3.0 %, respectively, and when the tube diameter is 15.88 mm, they increase by 9.8, 4.4, and 3.4 %, respectively. It can be seen that as the tube diameter is increased, the variation in performance increases with the changing pressure drop conditions. Table 6 shows the design values and performance results for the conditions considered optimal design points for the results in Figs. 8 and 9. Table 6 also shows the tube weight of the evaporator in each condition. At a pressure drop of 0.1 bar or less, the cooling capacity, COP, and SCP are approximately 995.5 W, 0.288, and 49.6 W/kg, respectively, regardless of the tube diameter of the heat exchanger. However, depending on the tube diameter condition of the heat exchanger, the mass of the tubes installed inside each evaporator varies from 15.4, 13.6, and 11.7 kg for tube diameters of 9.53, 12.7, and 15.88 mm, respectively. This means that the mass of the tubes in the heat exchanger required to achieve the same performance can increase by about 32 %, depending on the choice of the tube diameter. In general, when making a heat exchanger, the heat exchanger's price increases as the tubes' mass increases. Additionally, installing many smaller tubes will increase the cost of production compared to installing fewer larger tubes. From this point of view, it is recommended to select a tube diameter of 15.88 mm for the heat exchangers, both in terms of price and fabrication, when the pressure drop condition is 0.1 bar or less. Table 6 shows the optimal design points and



(a) Cooling capacity



(b) COP



(c) SCP

Fig. 10. Performance of chemisorption heat pump according to number of fins in reactor fin-tube.

system performance for pressure drop conditions of 1 bar or less. In this condition, the cooling capacity, COP, and SCP are about 1104 W, 0.295, and 49.5 W/kg, respectively, regardless of the tube diameter of the heat exchanger. For this pressure drop condition, it can be seen that the mass of the tubes in the evaporator is 15.4, 9.1, and 11.7 kg when the tube diameter of the heat exchanger is 9.53, 12.7, and 15.88 mm,

respectively. Therefore, when designing a system with a pressure drop of 1 bar or less, selecting a tube diameter of 12.7 mm for the heat exchangers is advantageous.

Fig. 10 shows the results of system performance according to the number of fins in the fin-tube. For Case 3 and Case 6, the optimal designs in Table 6, analysis was performed according to the number of fins in the fin-tube. As the number of fins in the fin-tube increases, the thermal conductivity in the fin-adsorbent section increases but the amount of adsorbent charged decreases. Therefore, increasing the number of fins has both positive and negative effects on system performance. As shown in Fig. 10(a), it can be seen that as the number of fins increases, the cooling capacity increases due to the increase in amount of adsorbent is significant. Although the specific value varies depending on each pressure drop condition, when the number of fins increases beyond the optimal number, it can be seen that the cooling capacity decreases despite the increase in the amount of adsorbent due to the decrease in thermal conductivity. As the number of fins with a relatively high density compared to the adsorbent increases, the heat capacity of the reactor's structure increases. Therefore, as described earlier, the amount of heat loss during the alternating operation process increases, and this reduces system COP as shown in Fig. 10(b). In the case of SCP, it can be seen that SCP increases as the number of fins increases. As the number of fins increases, the mass of the adsorbent decreases. The decrease in the mass of adsorbent due to the change in the number of fins is more dominant than the change in cooling capacity. Therefore, the result is as shown in Fig. 10(c). Based on the Fig. 10, it can be seen that the number of fins should be appropriately set depending on the design purpose.

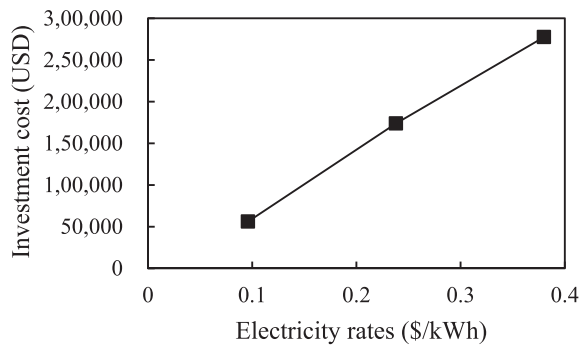
4.4. Economic analysis

An economic evaluation of a waste heat-driven heat pump was performed against a 100 kW electric-driven heat pump operating for 4380 h per year. An important factor to consider for economic evaluation is the cost of electricity. Electricity prices in major OECD countries were in range from 0.096 to 0.38 USD/kWh in 2022, with an average of 0.18 USD/kWh. Assuming a cooling COP of 3 for the electric heat pump, the electricity used per year is 146,000 kWh. The initial capital investment for a chemisorption heat pump system to have a net present value (NPV) of zero after five years, depending on the electricity price, is shown in Fig. 11(a). The cash flows are shown in Fig. 11(b) when the electricity price is 0.096, 0.238, and 0.38 USD/kWh and the initial capital investment is made so that the NPV is at the end of 5 years. Higher upfront costs are acceptable in areas where electricity is more expensive, and after five years, user can expect to see large annualized profits. Currently, 100 kW chemisorption heat pump systems are not commercially available, so it is not possible to determine specifically what the price will be, but compared to the price of a 100 kW electric heat pump, the price in Fig. 10(a) is likely to be achievable.

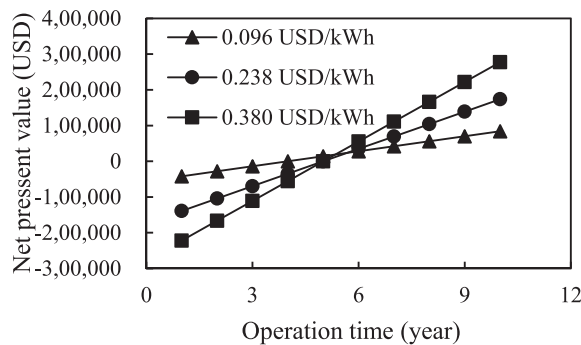
5. Conclusion

Interest in waste heat recovery systems to reduce carbon emissions is increasing, and accordingly, research on chemisorption heat pumps driven by heat sources is also being conducted around the world. In this study, unlike studies in existing literature, the performance of a chemisorption heat pump driven by a heat source as low as 40 °C was analyzed according to system design. The system performance was presented and analyzed according to the number of tubes inside the heat exchangers, the number of tubes in the reactor, the diameter of the tubes in heat exchanger, and the number of tube passes according to the differential pressure requirements of the heat exchanger and reactor. In addition, performance analysis was performed according to the number of fins in the reactor, and economic analysis under scale-up conditions was also conducted. For the performance analysis, a 1-D transient model was developed.

Here are the key findings.



(a) Initial investment cost for zero net present value (NPV) after 5 years operation



(b) NPV with respect to electricity price

Fig. 11. Economic evaluation of chemisorption heat pump.

- In most cases, the system performance improves as the number of tubes in the heat exchanger increases.
- Under some conditions, the number of heat exchanger passes decreases to meet the pressure drop condition as the number of tubes increases, which leads to a decrease in the tube side heat transfer coefficient and, consequently, a decrease in the system performance.
- When the differential pressure requirement of the secondary fluids of the heat exchangers and reactors is 0.1 bar, the optimal design conditions are when the outer diameter of the tubes in heat exchanger is 15.88 mm, the number of tubes in the evaporator and condenser is 20 and 40, respectively, and the number of fin-tubes inside the reactor is 180. Cooling capacity, COP, and SCP are 999.6 W, 0.288, and 49.8 W/kg, respectively.
- When the differential pressure requirement of the secondary fluids of the heat exchangers and reactors is 1 bar, the optimal design conditions are when the outer diameter of the tubes in heat exchanger is 12.7 mm, the number of tubes in the evaporator and condenser is 20 and 40, respectively, and the number of fin-tubes inside the reactor is 200. Cooling capacity, COP, and SCP are 1083.1 W, 0.291, and 48.6 W/kg, respectively.
- If the pressure drop requirement increases from 0.1 bar to 1 bar, the cooling capacity increases by approximately 8.4 %.
- There is a number of fins at which the cooling capacity is maximum, and the optimal number of fins per unit length (m) is 157 and 197 when the pressure drop requirement is 0.1 bar and 1 bar, respectively.
- The optimal design result varies depending on which items the user considers important among cooling capacity, COP, and SCP.
- Considering overall factors, the cooling capacity, COP, and SCP under optimal design conditions are 1100.6 W, 0.3, and 45.7 W/kg, respectively.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgments

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